

MICHAEL HARDY

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EDUCATION

Stanford University

Expected 2028

Ph.D., Data Science (Education)

Areas of Concentration: Measurement, AI, NLP, Evaluation, Statistical Methods, Psychometrics, Benchmarking, Assessment, Policy, Program Implementation, Data Use

GPA: 4.11/4.0

Fellowships: Stanford Graduate Fellow: Cisco Systems Fellow, MacArthur AI Measurement Science Fellow

University of California, Berkeley

2020

Master of Information and Data Science

Focus: Natural Language Processing | GPA: 3.93/4.0

Columbia University

2013

Master of Arts, Educational Leadership

GPA: 3.96/4.0 | First cohort member hired as principal and as area superintendent

Brigham Young University

2009

Bachelor of Science, Electrical Engineering (Magna cum laude)

Minors: Mathematics, Business Management, Linguistics | Study Abroad: Nanjing, China

GPA: 3.90/4.0 | Scholarships: Boeing, Fisher House, National Instruments, Jonsson, Shaeffer, Scholastic Gold

Relay Graduate School of Education

2017

National Principal Supervisors Academy Fellowship

Certification with distinction; invited to return as program advisor

PUBLICATIONS

Peer-Reviewed

Hardy, M. & Kim, Y. (2026, award pending). Knowledge without Wisdom: Measuring Misalignment between LLMs and Intended Impact. In *Proceedings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics.
<https://arxiv.org/abs/2603.00883>

Hardy, M., Reuel, A., Zhang, L., Casabianca, J. M., Truong, S. T., Dave, Y., Lee, H., Domingue, B. W., & Koyejo, S. (2026) AI Cartography: Mapping the Latent Landscape of AI Benchmark

- Ecosystems. In *Proceedings of the 43rd International Conference on Machine Learning*, (ICML 2026) Seoul, South Korea. PMLR 306, 2026. <https://arxiv.org/abs/2605.25272>
- Kim, Y., Hardy, M., Tey, J., Thille, C., & Piech, C. (2026) Interpretability from the Ground Up: Stakeholder-Centric Design of Automated Scoring in Educational Assessments. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics. <https://arxiv.org/abs/2511.17069>
- Truong, S. T.*, Tu, Y. *, Hardy, M.*, Reuel, A., Tang, Z., Burapachee, J., Perera, J. J., Uwakwe, C., Domingue, B. W., Haber, N., & Koyejo, S. (2025). Fantastic Bugs and Where to Find Them in AI Benchmarks. *The Thirty-Ninth Annual Conference on Neural Information Processing Systems* (Neurips) Datasets and Benchmarks Track. <https://openreview.net/forum?id=SlhLRh810S>
- Hardy, M. (2025). “All that Glitters”: Techniques for Evaluations with Unreliable Model and Human Annotations. In L. Chiruzzo, A. Ritter, & L. Wang (Eds.), *Findings of the Association for Computational Linguistics: NAACL 2025* (pp. 2250–2278). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.findings-naacl.120>
- Hardy, M. (2025). Measuring Teaching with LLMs. In J. Wilson, C. Ormerod, & M. Beiting Parrish (Eds.), *Proceedings of the Artificial Intelligence in Measurement and Education Conference* (AIME-Con): Full Papers (pp. 367–384). National Council on Measurement in Education (NCME). <https://aclanthology.org/2025.aimecon-main.40/>
- Hardy, M. (2020). Accounting for Accountability: An Analysis of the Tennessee Accountability System. Tennessee Department of Education. Peer-reviewed Report. <https://drive.google.com/file/d/1Oicyoye3oan9z-LuJYnuWiFNZLyh8gjs/view>

Keynotes and Invited Talks

- Hardy, M. (2024). The Work of Responsible AI in Education. 2024 Learnest Symposium for Ethical Artificial Intelligence in Education. Keynote conference talk.
- Hardy, M. (2016). Principals Helping Teachers Assess in Diverse Content Areas. Reidy Interactive Learning Series (IRLS). National Center for the Improvement of Educational Assessment (NCIEA). Invited conference talk.

Conference Presentations & Working Papers

- Hardy, M. (2026). Refactor and Verifactor Analyses: Direct Measures of the Assumptions of Unidimensionality. International Meeting of the Psychometric Society 2026 Annual Meeting, Seoul, South Korea. <https://arxiv.org/abs/2603.20938>
- Hardy, M. (2026). Refactor Analyses: Direct Measures of the Assumptions of Unidimensionality. National Council on Measurement in Education 2026 Annual Meeting, Los Angeles, CA. <https://arxiv.org/abs/2603.20938>
- Hardy, M. & Domingue, B. (2026). Efficient Detection of Bad Items with Applications to AI Benchmarks. National Council on Measurement in Education 2026 Annual Meeting, Los Angeles, CA. <https://arxiv.org/abs/2603.24999>

- Hardy, M. (2026). Autoscoring Anticlimax: Meta-analytic Understanding of Poor LLM Measurement. National Council on Measurement in Education 2026 Annual Meeting, Los Angeles, CA.
<https://arxiv.org/abs/2603.04820>
- Hardy, M. (2024). Rater Effects vs Robo-Effects: Profiling Automated Classroom Evaluation. National Council on Measurement in Education 2024 Annual Meeting, Philadelphia, PA.
- Hardy, M. (2023). Classroom Heartbeats: Improving Measurement of Classroom Instruction with Natural Language. Stanford Data Science Conference. **Best Poster Award, CEPA.**

Preprints & Working Papers

- Hardy, M. (2024). “All that Glitters”: Approaches to Evaluations with Unreliable Model and Human Annotations (No. arXiv:2411.15634). arXiv. <https://doi.org/10.48550/arXiv.2411.15634>
- Hardy, M. (2021). Toward Educator-focused Automated Scoring Systems for Reading and Writing (No. arXiv:2112.11973). arXiv. <https://doi.org/10.48550/arXiv.2112.11973>

COURSES TAUGHT

- CS 321M: AI Measurement Science (Instructor, Spring 2026, Stanford University)

PROFESSIONAL EXPERIENCE

Executive Director; Senior Advisor/Director Emeritus 2023–2025

Learnest (Non-profit), Part-time, Pro Bono

- Led all aspects of a non-profit dedicated to understanding and influencing the ethical use of AI in education, including programming for a fellowship experience
- Inaugurated initial fellowship cohort; delivered keynote address at AI Ethics symposium

Head of Academics (Founding Team) 2023–2024

Atypical AI, Education Technology Startup

- Led measurement, curricula, educator resources, instructional content creation, research, pedagogy, evaluation, assessment, and alignment
- Set visions for measurement of learning, data, and pedagogy; built initial models to secure investor funding

Chief Strategy and Data Officer 2019–2021

Tennessee Department of Education, Nashville, TN

- Led divisions of assessment, data analytics, research, evaluation, data science, data governance, and organizational strategy for Tennessee before and through COVID-19
- Restored confidence in TN assessment program during pandemic; three consecutive years of flawless administration and reporting; led nation in measuring COVID learning loss
- Saved \$60MM over 5 years via contract writing and negotiation
- Designed 5-year evaluation reform plan (peer-reviewed); launched roadmap

- Set vision, wrote contract, and procured statewide universal assessments for all TN PK–3 students to measure learning loss
- Won grant and launched TN Innovative Assessment program: 99% of school districts adopted new tests
- Improved all previous assessment and data reporting products and services with a 25% reduction in FTE staff

Senior Vice President of Schools (Area Superintendent) & Vice President of School Analytics 2015–2019

IDEA Public Schools, San Antonio, TX

- As SVP: Directly managed 9 principals leading schools serving 3,500 students and 250 team members in the fastest growing charter management organization in the nation
- Led strongest new school launches in organization; greatest growth in academic performance K–2 (48% on-grade increase), educator retention (31% increase), student & teacher attendance (1.5% increase), and family re-enrollment (5% increase)
- 100% of managed schools earned 100% of all state distinctions; highest portfolio performance in subsequent years
- 100% of schools received A/A+ from Children at Risk 2017 Evaluation; highest average reading and math impact per CREDO study
- As VP of Analytics: Principal Data Scientist; led data analytics and strategy program for 28,000 students, staff retention and evaluations for 1,700 educators across 65 schools
- Work recognized by NCIEA; invited to present at 2016 Reidy Interactive Lecture Series
- Most impactful business partner as voted by principals and vice presidents (2018, 2019)
- Team of the Year for national organization
- Led school turnaround efforts moving lowest-performing school out of risk of unacceptable accountability rating

Director of Leader Development

2014–2015

IDEA Public Schools, San Antonio, TX

- Provided executive coaching to 17 principals and principals-in-residence; designed & executed year-long training for 25 assistant principals and 50 teacher leaders
- 100% of principals-in-residence coached promoted to principalship; multiple have risen to senior leadership
- Leaders in coaching portfolio had greatest growth in student impact based on internal and state assessments

Principal & Administrator

2011–2014

IDEA Frontier College Preparatory, Brownsville, TX

- Led 800 middle and high school students (90% economically disadvantaged, 98% minority) to 90% passing rate on state assessments—higher than state and local districts
- Increased teacher retention rate by 31%; highest performing full-scale school in state and AP exams in district, even with 70% of teachers having <1 year experience
- 100% of graduates matriculated into college; 72% first-generation college students

- Awarded top 1% national ranking: US News & World Report (#8 TX, #46 USA); Washington Post Most Challenging Schools (#8 TX, #22 USA)
- Texas Charter Achievement Award (each year); State Comptroller's Award for Fiscal Responsibility; Secondary School of the Year for IDEA Public Schools

Teacher and Program Administrator

2009–2011

Pace High School, Brownsville, TX (Teach for America)

- Taught STEM courses to 98% economically disadvantaged and 97% minority students
- Closed the achievement gap between high- and low-income students on state science exam
- Co-authored, won, and managed execution of competitive \$6M school transformation grant
- Founded largest FTC robotics team in nation; won Texas State Championship FTC Robotics as coach

Finance Analyst

2009

Beneficial Financial Group

- Performed in-depth research and analysis on structured assets during financial crisis
- 100% of portfolio credit risk recommendations accepted by senior leadership; offered full-time position after first presentation

Engineering Technician

2007–2008

Sandia National Laboratories

- Constructed robotic lab equipment for automated, more precise electrochemical experimentation—first of its kind built from off-the-shelf parts
- Saved over \$100,000 in machinery; increased efficiency in experimentation
- Authored report on Strategic Partnership Programs for the Department of Energy; earned SPOT Award for performance excellence

AWARDS & DISTINCTIONS

- Stanford Graduate Fellow, Cisco Systems Fellow, Stanford University
- MacArthur AI Measurement Science Fellow
- Best Poster Award, Center for Education Policy Analysis (CEPA), Stanford Data Science Conference
- Keynote Address, AI Ethics Symposium
- As Teacher:
 - Texas Educator of the Year 2012 (HEB Secondary Rising Star Category—TX has the most public-school teachers)
 - Texas State Champions, FTC Robotics (Coach)
- South Texas Regional Educator of the Year 2012
- As Principal of IDEA Frontier College Preparatory:
 - US News & World Report Top 1% National Ranking (#8 Texas, #46 USA)
 - Washington Post Most Challenging Schools (#8 Texas, #22 USA)
 - Texas Charter Achievement Award (multiple years)
 - State Comptroller's Award for Fiscal Responsibility
- As District Leader
 - NCIEA Recognition; Invited Presenter, 2016 Reidy Interactive Lecture Series (RILS)

- Team of the Year, IDEA Public Schools (National Organization)
- Sandia National Laboratories SPOT Award for Performance Excellence
- Boeing, Fisher House, National Instruments, Jonsson, Shaeffer, and Scholastic Gold Scholarships

TECHNICAL SKILLS & EXPERTISE

Measurement Science

Psychometric and Measurement Models, Validity Analysis, Metric Construction, Item Response Theory, Generalizability Theory, Measure Theory, Nonparametric and Bayesian Methods, Causal Inference, Factorial Analyses

AI Development & Natural Language Processing

Deep and Machine Learning Models. Tools: PyTorch, TensorFlow, Keras, Python, R, including distributed learning and deep learning methods. Current state-of-the-art for models that evaluate the quality of human teaching.

Deep Learning: Expertise building custom AI components including optimizers, loss functions, multi-input/multi-output models, learning rate functions, embedding, novel NLP dataset extension, transformer attention models. Developed Automated Essay Grader achieving state-of-the-art results for essay scoring.

SERVICE & PROFESSIONAL MEMBERSHIPS

- Executive Director, Learnest (non-profit for ethical AI in education), 2023–2025
- Teach for America Corps Member, 2009–2011

PERSONAL

Mike is an award-winning research data scientist and educator. He has advanced degrees from Columbia and Berkeley and is currently pursuing a PhD at Stanford in data science and measurement where he is a graduate fellow. His research includes measuring and differentiating between natural and artificial learning and assessment. Previously Chief Strategy and Data Officer with over seven years as a senior executive, he has a track record of excellence at every level of public education, from a Educator of the Year Award in Texas to launching innovative work as Chief Strategy Data Officer for TNDOE, with a track record of success in each role. He is an expert in teaching, learning, curriculum, and assessment, using those skills to lead his high school to top 1% national rankings by US News and World Report and Washington Post's Most Challenging Schools. He plays Wordle every morning and loves fostering pets—he has fostered over 70 kittens and cats and has named them all scientists and mathematicians. And he is not above binge watching total garbage on Netflix.